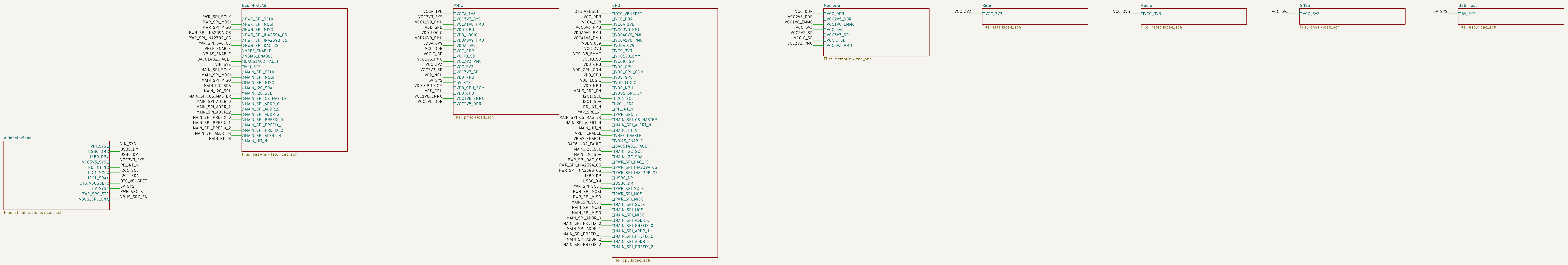
Alimentazione
VIN_SYS
USB0_DM
USB0_DP
VCC3V3_SYS
PD_INT_N
I2C1_SCL
I2C1_SDA
OTG_VBUSDET
5V_SYS
PWR_SRC_ST
VBUS_SRC_EN
VIN_SYS
USB0_DM
USB0_DP
VCC3V3_SYS
PD_INT_N
I2C1_SCL
I2C1_SDA
OTG_VBUSDET
5V_SYS
PWR_SRC_ST
VBUS_SRC_EN
File: alimentazione.kicad_sch
Bus MIKILAB
PWR_SPI_SCLK
PWR_SPI_MOSI
PWR_SPI_MISO
PWR_SPI_INA239A_CS
PWR_SPI_INA239B_CS
PWR_SPI_DAC_CS
VREF_ENABLE
VBIAS_ENABLE
DAC61402_FAULT
VIN_SYS
MAIN_SPI_SCLK
MAIN_SPI_MOSI
MAIN_SPI_MISO
MAIN_I2C_SDA
MAIN_I2C_SCL
MAIN_SPI_CS_MASTER
MAIN_SPI_ADDR_0
MAIN_SPI_ADDR_1
MAIN_SPI_ADDR_2
MAIN_SPI_PREFIX_0
MAIN_SPI_PREFIX_1
MAIN_SPI_PREFIX_2
MAIN_SPI_ALERT_N
MAIN_HIT_N
PWR_SPI_SCLK
PWR_SPI_MOSI
PWR_SPI_MISO
PWR_SPI_INA239A_CS
PWR_SPI_INA239B_CS
PWR_SPI_DAC_CS
VREF_ENABLE
VBIAS_ENABLE
DAC61402_FAULT
VIN_SYS
MAIN_SPI_SCLK
MAIN_SPI_MOSI
MAIN_SPI_MISO
MAIN_I2C_SDA
MAIN_I2C_SCL
MAIN_SPI_CS_MASTER
MAIN_SPI_ADDR_0
MAIN_SPI_ADDR_1
MAIN_SPI_ADDR_2
MAIN_SPI_PREFIX_0
MAIN_SPI_PREFIX_1
MAIN_SPI_PREFIX_2
MAIN_SPI_ALERT_N
MAIN_HIT_N
File: bus-mikilab.kicad_sch
PMIC
VCCA_1V8
VCC3V3_SYS
VCCA1V8_PMU
VDD_GPU
VDD_LOGIC
VDDA0V9_PMU
VDDA_0V9
VCC_DDR
VCCIO_SD
VCC3V3_PMU
VCC_3V3
VCC3V3_SD
VDD_NPU
5V_SYS
VDD_CPU_COM
VDD_CPU
VCC1V8_EMMC
VCC2V5_DDR
VCCA_1V8
VCC3V3_SYS
VCCA1V8_PMU
VDD_GPU
VDD_LOGIC
VDDA0V9_PMU
VDDA_0V9
VCC_DDR
VCCIO_SD
VCC3V3_PMU
VCC_3V3
VCC3V3_SD
VDD_NPU
5V_SYS
VDD_CPU_COM
VDD_CPU
VCC1V8_EMMC
VCC2V5_DDR
File: pmic.kicad_sch
CPU
OTG_VBUSDET
VCC_DDR
VCCA_1V8
VCC3V3_PMU
VDDA0V9_PMU
VCCA1V8_PMU
VDDA_0V9
VCC_3V3
VCC1V8_EMMC
VCCIO_SD
VDD_CPU
VDD_CPU_COM
VDD_GPU
VDD_LOGIC
VDD_NPU
VBUS_SRC_EN
I2C1_SCL
I2C1_SDA
PD_INT_N
PWR_SRC_ST
MAIN_SPI_CS_MASTER
MAIN_SPI_ALERT_N
MAIN_HIT_N
VREF_ENABLE
VBIAS_ENABLE
DAC61402_FAULT
MAIN_I2C_SCL
MAIN_I2C_SDA
PWR_SPI_DAC_CS
PWR_SPI_INA239A_CS
PWR_SPI_INA239B_CS
USB0_DP
USB0_DM
PWR_SPI_SCLK
PWR_SPI_MOSI
PWR_SPI_MISO
MAIN_SPI_SCLK
MAIN_SPI_MOSI
MAIN_SPI_MISO
MAIN_SPI_ADDR_0
MAIN_SPI_PREFIX_0
MAIN_SPI_ADDR_1
MAIN_SPI_PREFIX_1
MAIN_SPI_ADDR_2
MAIN_SPI_PREFIX_2
OTG_VBUSDET
VCC_DDR
VCCA_1V8
VCC3V3_PMU
VDDA0V9_PMU
VCCA1V8_PMU
VDDA_0V9
VCC_3V3
VCC1V8_EMMC
VCCIO_SD
VDD_CPU
VDD_CPU_COM
VDD_GPU
VDD_LOGIC
VDD_NPU
VBUS_SRC_EN
I2C1_SCL
I2C1_SDA
PD_INT_N
PWR_SRC_ST
MAIN_SPI_CS_MASTER
MAIN_SPI_ALERT_N
MAIN_HIT_N
VREF_ENABLE
VBIAS_ENABLE
DAC61402_FAULT
MAIN_I2C_SCL
MAIN_I2C_SDA
PWR_SPI_DAC_CS
PWR_SPI_INA239A_CS
PWR_SPI_INA239B_CS
USB0_DP
USB0_DM
PWR_SPI_SCLK
PWR_SPI_MOSI
PWR_SPI_MISO
MAIN_SPI_SCLK
MAIN_SPI_MOSI
MAIN_SPI_MISO
MAIN_SPI_ADDR_0
MAIN_SPI_PREFIX_0
MAIN_SPI_ADDR_1
MAIN_SPI_PREFIX_1
MAIN_SPI_ADDR_2
MAIN_SPI_PREFIX_2
File: cpu.kicad_sch
Memorie
VCC_DDR
VCC2V5_DDR
VCC1V8_EMMC
VCC_3V3
VCC3V3_SD
VCCIO_SD
VCC3V3_PMU
VCC_DDR
VCC2V5_DDR
VCC1V8_EMMC
VCC_3V3
VCC3V3_SD
VCCIO_SD
VCC3V3_PMU
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Rete
VCC_3V3
VCC_3V3
File: rete.kicad_sch
Radio
VCC_3V3
VCC_3V3
File: radio.kicad_sch
GNSS
VCC_3V3
VCC_3V3
File: gnss.kicad_sch
USB host
5V_SYS
5V_SYS
File: usb.kicad_sch

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CPU: Rockchip RK3568 (FCBGA636L 19x19 mm, passo 0.65 mm, LCSC C2938930). Ball map: Rockchip RK3568 PinOut V1.1, verificata sulla tabella 2-1 del datasheet V1.2 (636/636). Serie 3xx.

Domini IO: PMUIO1/PMUIO2 = VCC3V3_PMU (3.3 V), VCCIO2 = VCC1V8_EMMC (1.8 V, FLASH_VOL_SEL alto), VCCIO3 = VCCIO_SD, VCCIO4-7 = VCC_3V3 (3.3 V), VCCIO1 non usato (flottante, come da Rockchip).

PHY non usati (USB2 HOST2/3, MULTI_PHY, PCIe3, MIPI CSI/DSI, HDMI, eDP): alimentazioni a VSS e segnali flottanti, secondo ‘RK3568 Processing instructions for unused pins’. VDD_NPU su BUCK4 dell’RK809-5.

DDR4 a 16 bit sul canale A del PHY (DQ0-15_A, nomi DDR4_DQL/DQU_A del pinout Rockchip); canale B, ECC, CS1n, ODT1, BG1 non usati. Mappa completa in docs/pinmap-rk3568.md.

U301A RK3568 DDR PHY
U301C RK3568 VSS
U301D RK3568 AVSS
PMUIO1-2 / OSC / SARADC
VCCIO5-7
VCCIO1-4 (EMMC, SDMMC0, GMAC0)
ALIMENTAZIONI
USB / PCIe / DATA
VIDEO (non usati)
SW301 RESET
SW302 RECOVERY
J301 UART2 debug
Y301 24MHz
TPD4E05U06DQA

24 MHz (Hardware Design Guide RK3568 fig. 2-1 / reference AIoT): 22R serie su XOUT24M, 1M tra XIN e XOUT, 18 pF per lato (da adattare al CL del quarzo, tolleranza 20 ppm).

nPOR (AH27, dominio PMUIO1): rete RESETn con 100 nF vicino alla sfera, RESETB dell’RK809-5 (open drain, 10k a VCC3V3_PMU sul foglio PMIC), TSADC_SHUT_M0 con 0R e tasto RESET.

Il tasto RESET tira giu’ RESETn: l’RK809-5 rifa’ la sequenza di accensione (Hardware Design Guide, par. Reset). FLASH_VOL_SEL (GPIO0_A7) 10k a VCC3V3_PMU: VCCIO2 a 1.8 V.

Recovery (ex FEL): SARADC_VIN0 (B27) con 10k a VCCA_1V8 (obbligatorio, Rockchip) e tasto a GND: tenuto premuto all’accensione il loader entra in modalita’ Recovery/rockusb.

J301 console UART2_M0 (GPIO0_D1 TX, GPIO0_D0 RX, PMUIO2 3.3 V): 1 = TX del SoC, 2 = RX del SoC, 3 = GND. 1500000 baud (default Rockchip).

I2C0 (GPIO0_B1/B2, PMUIO2): RK809-5 0x20 e TCS4525 0x1C, pull-up 2.2k a VCC3V3_PMU. I2C1 (GPIO0_B3/B4): FUSB302B 0x22, 2.2k a VCC3V3_PMU.

Open drain verso i GPIO PMU: PMIC_INT_L, PD_INT_N (FUSB302B), USB1_OC_N (TPS2553), PWR_SRC_ST (TPS2121) con 10k a VCC3V3_PMU. SD_DET 10k sul foglio Memorie.

USB3_OTG0_DP/DM (porta di flash rkdeveloptool, USB-C di alimentazione) con 2.2R in serie. USB3_HOST1_DP/DM verso la USB-C host (2.2R sul foglio USB host).

USB3_AVDD_0V9/1V8 da VDDA_0V9/VCCA_1V8 con ferrite 120R@100MHz (reference AIoT), USB3_AVDD_3V3 da VCC_3V3. USB2_AVDD (HOST2/3 non usati) a VSS.

Disaccoppiamento secondo il reference Rockchip RK3568 AIoT V1.1: 100 nF/1 nF in 0201 e 4.7 uF/1 uF in 0402 sotto il package, 22 uF/10 uF 0603 vicino.

VDD_CPU >= 135 uF in totale con le uscite del TCS4525 (Hardware Design Guide 2.2.2.4), un 22 uF DNP di riserva. VDD_LOGIC/GPU/NPU >= 90 uF con le uscite dell’RK809-5.

C315 DNP: 22 uF di riserva su VDD_CPU (Hardware Design Guide). VCCIO4 ha due 100 nF (RGMII), VCCIO5/VCCIO6 due sfere ciascuno.

DDR_RZQ (H7): 120R 1% a GND per DDR4 (Hardware Design Guide 2.1.7.2; l’appunto diceva 240R, che vale per lo ZQ della DDR4). DDR_VREFOUT (P8): 1 nF a VCC_DDR e 1 nF a GND vicino al SoC.

BUS_POWER su SPI3_M1 (VCCIO7): CLK GPIO4_C2, MOSI GPIO4_C3, MISO GPIO4_C5, CS0 INA239A GPIO4_C6, CS1 INA239B GPIO4_D1; CS del DAC su GPIO4_C4 con cs-gpios (SPI3 ha solo due CS hardware).

MAIN su SPI2_M1 (VCCIO5): CLK GPIO3_A0, MOSI GPIO2_D6, MISO GPIO2_D7, CS0 = CS_MASTER GPIO2_D5; CS1 (GPIO2_D4) = EEPROM MAC locale. 33R serie su CLK/MOSI/MISO verso i connettori.

I2C del bus su I2C3_M1 (GPIO3_B5/B6). ADDR/PREFIX con 100R serie. Linee lente (VREF/VBIAS_ENABLE, FAULT, ALERT#, HIT#, PPS, EXTINT) su GPIO diretti: niente TCA6416A.

Livelli di riposo: CS e CS_MASTER alti (10k), VREF_ENABLE basso (10k a GND), VBIAS_ENABLE basso (100k a GND), ALERT# e HIT# 4.7k, I2C del bus 2.2k, FAULT 10k (tutti a VCC_3V3).

ESD bassa capacita’ TPD4E05U06DQA (0.5 pF, 5.5 V) sulle linee logiche di J1/J2, vicino ai connettori. Nessuna protezione su V_BIAS e riferimenti.

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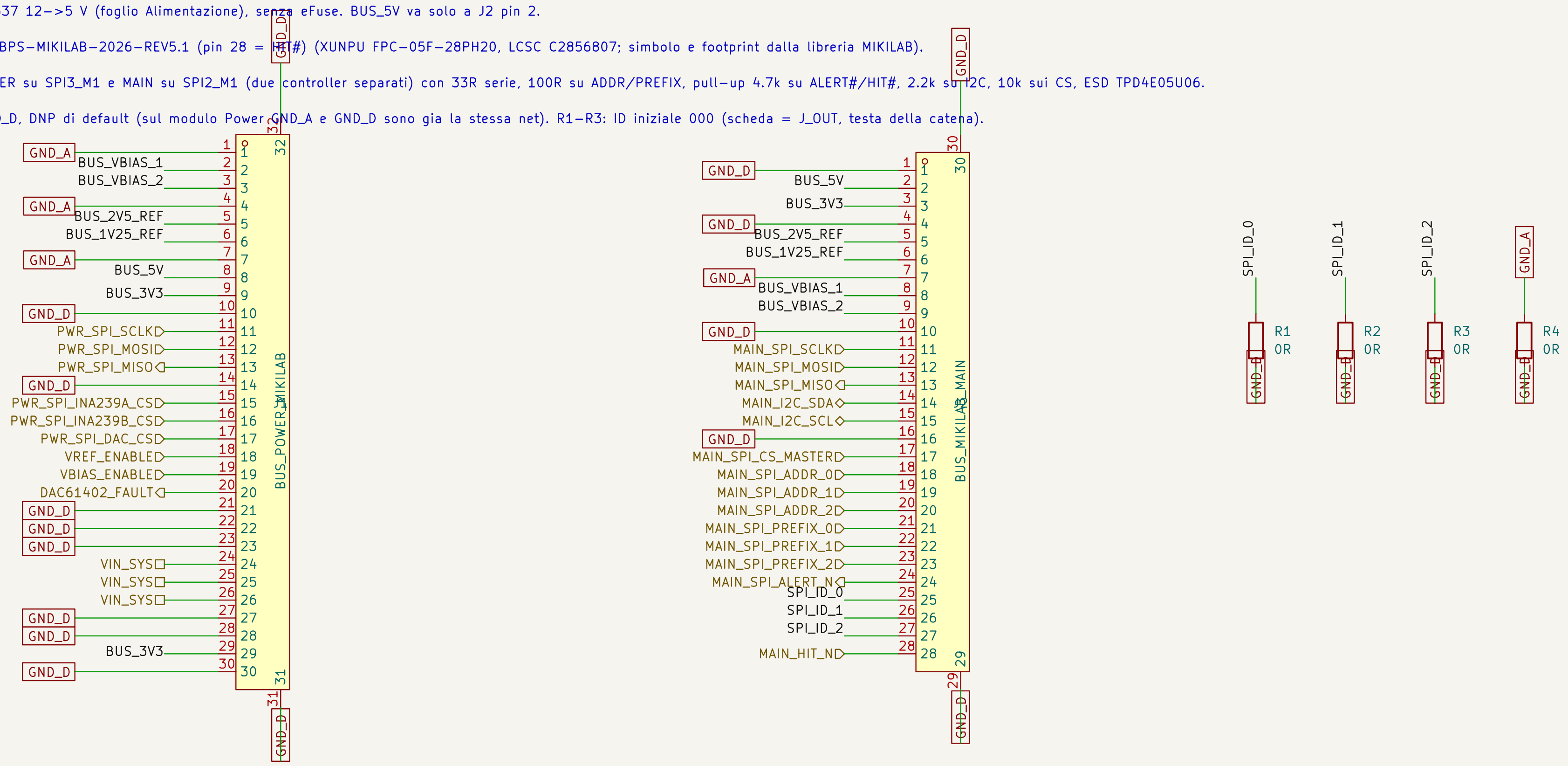
J1 BUS_POWER 30 pin (Hirose FH12-30S): pin 1-22 come J3 del modulo Power, 24-26 VIN_SYS 12 V, 29 secondo 3V3. Rail BUS_* passanti verso J2, non collegati ai rail della scheda.

VIN_SYS -> buck TPS56637 12->5 V (foglio Alimentazione), senza eFuse. BUS_5V va solo a J2 pin 2.

J2 MAIN 28 pin secondo BPS-MIKILAB-2026-REV5.1 (pin 28 = HIT#) (XUNPU FPC-05F-28PH20, LCSC C2856807; simbolo e footprint dalla libreria MIKILAB).

Sul foglio CPU: BUS_POWER su SPI3_M1 e MAIN su SPI2_M1 (due controller separati) con 33R serie, 100R su ADDR/PREFIX, pull-up 4.7k su ALERT#/HIT#, 2.2k su I2C, 10k sui CS, ESD TPD4E05U06.

R4 0R: ponte GND_A-GND_D, DNP di default (sul modulo Power GND_A e GND_D sono gia la stessa net). R1-R3: ID iniziale 000 (scheda = J_OUT, testa della catena).



J1
BUS_POWER_MIKILAB
GND_A
BUS_VBIAS_1
BUS_VBIAS_2
BUS_2V5_REF
BUS_1V25_REF
BUS_5V
BUS_3V3
GND_D
PWR_SPI_SCLK
PWR_SPI_MOSI
PWR_SPI_MISO
PWR_SPI_INA239A_CS
PWR_SPI_INA239B_CS
PWR_SPI_DAC_CS
VREF_ENABLE
VBIAS_ENABLE
DAC61402_FAULT
VIN_SYS
J2
BUS_MIKILAB_MAIN
MAIN_SPI_SCLK
MAIN_SPI_MOSI
MAIN_SPI_MISO
MAIN_I2C_SDA
MAIN_I2C_SCL
MAIN_SPI_CS_MASTER
MAIN_SPI_ADDR_0
MAIN_SPI_ADDR_1
MAIN_SPI_ADDR_2
MAIN_SPI_PREFIX_0
MAIN_SPI_PREFIX_1
MAIN_SPI_PREFIX_2
MAIN_SPI_ALERT_N
SPI_ID_0
SPI_ID_1
SPI_ID_2
MAIN_HIT_N
R1
0R
R2
0R
R3
0R
R4
0R

Sheet: /Bus MIKILAB/
File: bus-mikilab.kicad_sch

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Size: A3
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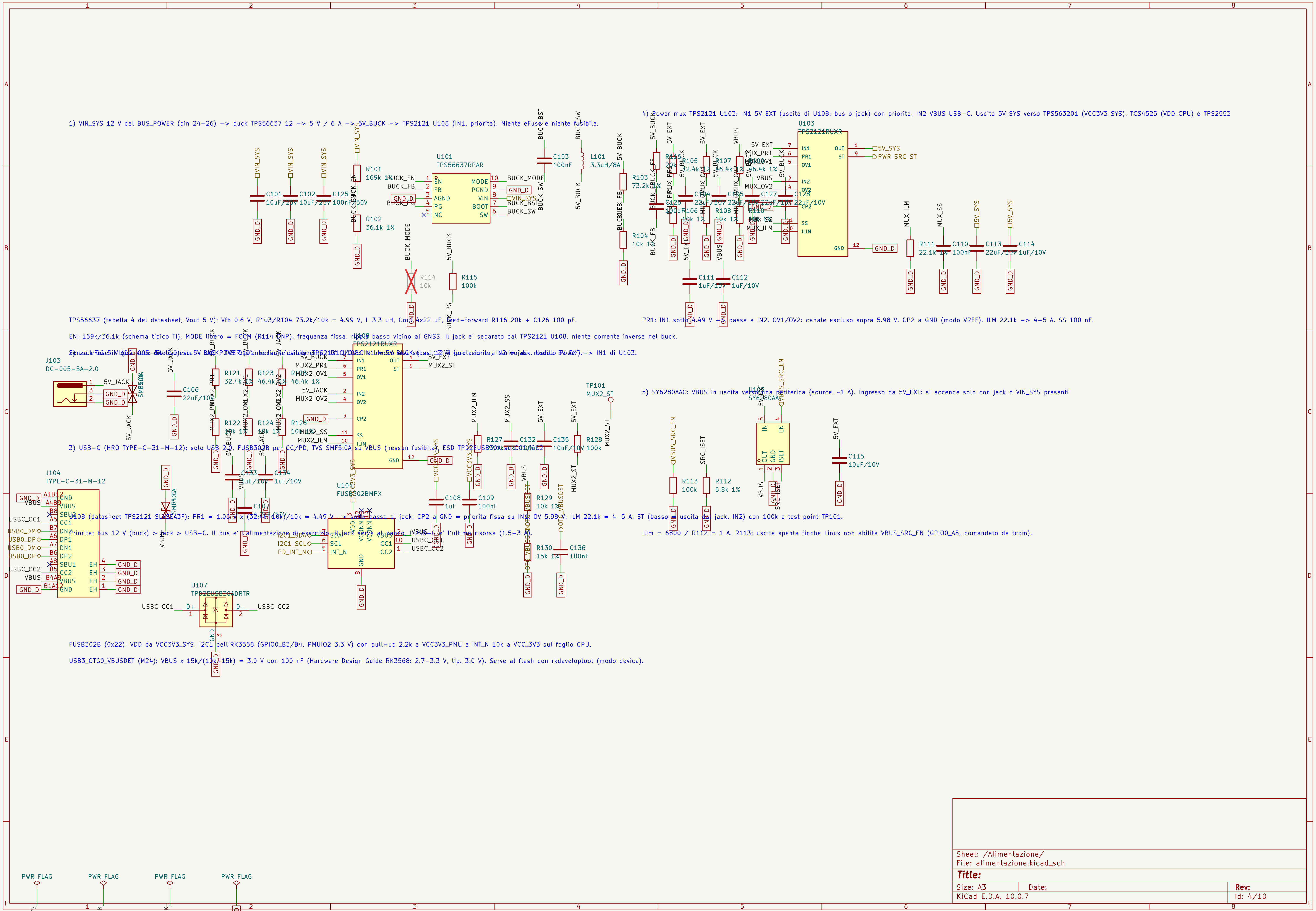
U101
TPS56637RPAR
BUCK_EN
BUCK_FB
BUCK_PG
BUCK_MODE
BUCK_BST
BUCK_SW
VIN_SYS
GND_D
C101 10uF/25V
C102 10uF/25V
C125 100nF/50V
R101 169k 1%
R102 36.1k 1%
C103 100nF
L101 3.3uH/8A
R103 73.2k 1%
R104 10k 1%
R114 10k
R115 100k
R116 20k
C126 100pF
U103
TPS2121RUXR
5V_EXT
MUX_PR1
MUX_OV1
VBUS
MUX_OV2
MUX_SS
MUX_ILM
5V_SYS
PWR_SRC_ST
R105 32.4k 1%
R106 10k 1%
R107 46.4k 1%
R108 10k 1%
R109 46.4k 1%
R110 10k 1%
C104 22uF/10V
C105 22uF/10V
C127 22uF/10V
C128 22uF/10V
C111 1uF/10V
C112 1uF/10V
R111 22.1k 1%
C110 100nF
C113 22uF/10V
C114 1uF/10V
J103
DC-005-5A-2.0
5V_JACK
D101 SMF5.0A
C106 22uF/10V
U108
TPS2121RUXR
5V_BUCK
MUX2_PR1
MUX2_OV1
MUX2_OV2
MUX2_SS
MUX2_ILM
MUX2_ST
R121 32.4k 1%
R122 10k 1%
R123 46.4k 1%
R124 10k 1%
R125 46.4k 1%
R126 10k 1%
C133 1uF/10V
C134 1uF/10V
R127 22.1k 1%
C132 100nF
C135 10uF/10V
R128 100k
TP101
MUX2_ST
J104
TYPE-C-31-M-12
USBC_CC1
USBC_CC2
USB0_DM
USB0_DP
D102 SMF5.0A
C107 4.7uF/10V
U104
FUSB302BMPX
I2C1_SDA
I2C1_SCL
PD_INT_N
VCC3V3_SYS
C108 1uF
C109 100nF
OTG_VBUSDET
R129 10k 1%
R130 15k 1%
C136 100nF
U107
TPD2EUSB30ADRTR
U105
SY6280AAC
VBUS_SRC_EN
SRC_ISET
R113 100k
R112 6.8k 1%
C115 10uF/10V
PWR_FLAG

1) VIN_SYS 12 V dal BUS_POWER (pin 24–26) –> buck TPS56637 12 –> 5 V / 6 A –> 5V_BUCK –> TPS2121 U108 (IN1, priorita). Niente eFuse e niente fusibile.

TPS56637 (tabella 4 del datasheet, Vout 5 V): Vfb 0.6 V, R103/R104 73.2k/10k = 4.99 V, L 3.3 uH, Cout 4x22 uF, feed–forward R116 20k + C126 100 pF.

EN: 169k/36.1k (schema tipico TI). MODE libero = FCCM (R114 DNP): frequenza fissa, ripple basso vicino al GNSS. Il jack e' separato dal TPS2121 U108, niente corrente inversa nel buck.

2) Jack DC 5 V (DC–005–5A–2.0) su 5V_JACK, TVS D101, nessun fusibile. TPS2121 U108: IN1 = 5V_BUCK (bus 12 V) con priorita, IN2 = jack. Uscita 5V_EXT –> IN1 di U103.

Senza eFuse il [illegible]

3) USB–C (HRO TYPE–C–31–M–12): solo USB 2.0, FUSB302B per CC/PD, TVS SMF5.0A su VBUS (nessun fusibile), ESD TPD2EUSB30 su CC1/CC2

U108 (datasheet TPS2121 SLVSEA3F): PR1 = 1.06 V x (32.4k+10k)/10k = 4.49 V –> sotto passa al jack; CP2 a GND = priorita fissa su IN1; OV 5.98 V; ILM 22.1k = 4–5 A; ST (basso = uscita dal jack, IN2) con 100k e test point TP101.

Priorita: bus 12 V (buck) > jack > USB–C. Il bus e' l'alimentazione di esercizio, il jack serve al banco, USB–C e' l'ultima risorsa (1.5–3 A).

4) Power mux TPS2121 U103: IN1 5V_EXT (uscita di U108: bus o jack) con priorita, IN2 VBUS USB–C. Uscita 5V_SYS verso TPS563201 (VCC3V3_SYS), TCS4525 (VDD_CPU) e TPS2553

PR1: IN1 sotto 4.49 V –> passa a IN2. OV1/OV2: canale escluso sopra 5.98 V. CP2 a GND (modo VREF). ILM 22.1k –> 4–5 A. SS 100 nF.

5) SY6280AAC: VBUS in uscita verso una periferica (source, ~1 A). Ingresso da 5V_EXT: si accende solo con jack o VIN_SYS presenti

Ilim = 6800 / R112 = 1 A. R113: uscita spenta finche Linux non abilita VBUS_SRC_EN (GPIO0_A5, comandato da tcpm).

FUSB302B (0x22): VDD da VCC3V3_SYS, I2C1 dell'RK3568 (GPIO0_B3/B4, PMUIO2 3.3 V) con pull–up 2.2k a VCC3V3_PMU e INT_N 10k a VCC_3V3 sul foglio CPU.

USB3_OTG0_VBUSDET (M24): VBUS x 15k/(10k+15k) = 3.0 V con 100 nF (Hardware Design Guide RK3568: 2.7–3.3 V, tip. 3.0 V). Serve al flash con rkdeveloptool (modo device).

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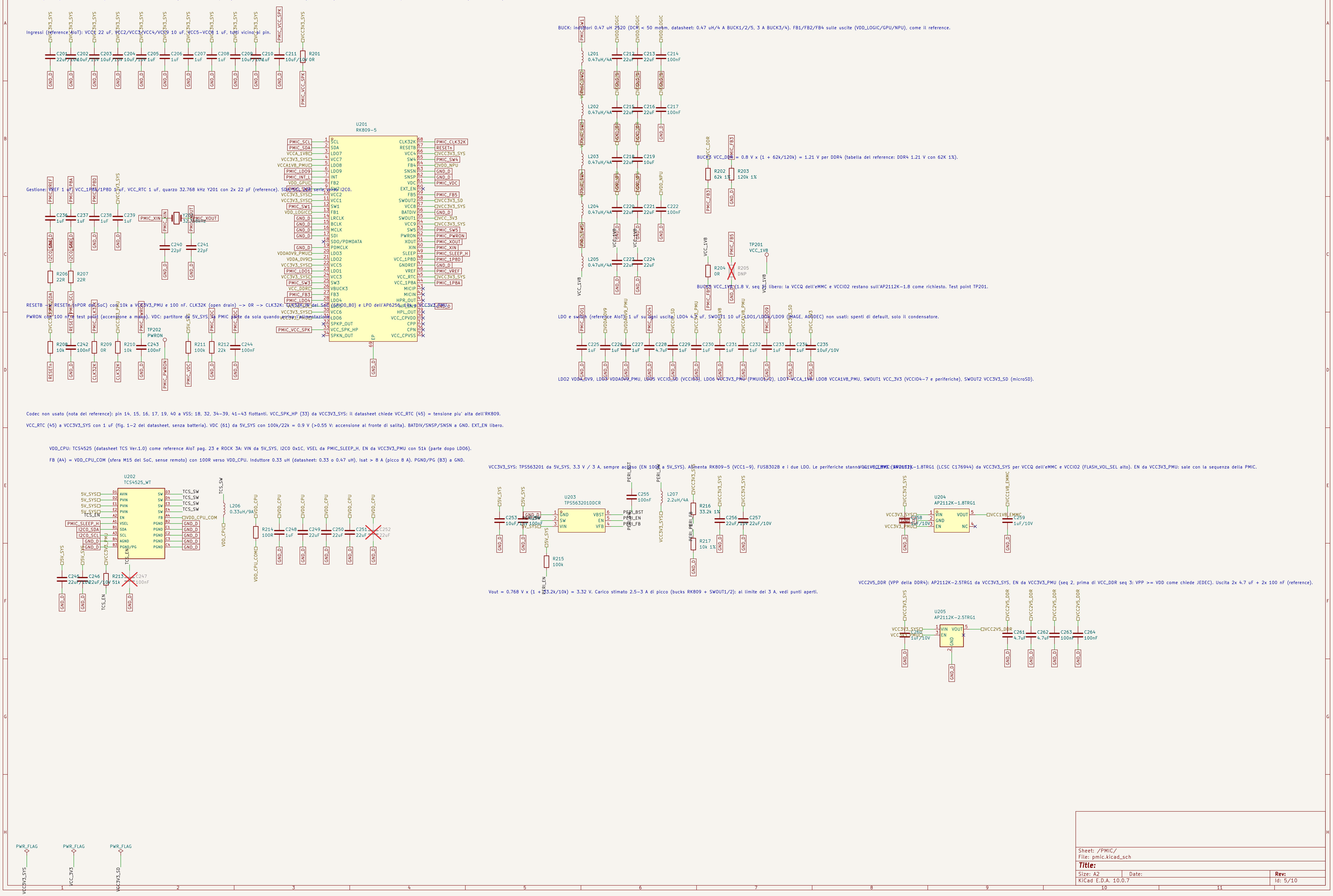
PMIC RK809-5 (datasheet Rockchip RK809 Rev 2.3; schema come Rockchip RK3568 AIoT REF V1.1 pag. 22 e Radxa ROCK 3A): ingressi VCC1-VCC9 da VCC3V3_SYS, I2C0 0x20. Serie 2xx.

Default RK809-5 (tabella 4-4): BUCK1 0.9 V seq 1, BUCK2 0.9 V seq 2, BUCK3 da partitore seq 3, BUCK4 SPENTO (0.5 V), BUCK5 1.8 V seq 2, LDO2/3 0.9 V seq 1, LDO5 3.3 V seq 4, LDO6 3.3 V seq 2, LDO7/8 1.8 V seq 2, SWOUT1/2 seq 4.

BUCK4 parte spento a 0.5 V: non puo' essere il 3.3 V di sistema (resterebbe tutto spento). Il 3.3 V resta sul TPS563201 (VCC3V3_SYS) come nel reference Rockchip; BUCK4 alimenta VDD_NPU (come ROCK 3A).


U201
RK809-5
U202
TCS4525_WT
U203
TPS563201DDCR
U204
AP2112K-1.8TRG1
U205
AP2112K-2.5TRG1
Y201
32.768kHz
L201 0.47uH/4A
L202 0.47uH/4A
L203 0.47uH/4A
L204 0.47uH/4A
L205 0.47uH/4A
L206 0.33uH/9A
L207 2.2uH/4A
R202 62k 1%
R203 120k 1%
R204 0R
R205 DNP
R214 100R
R215 100k
R216 33.2k 1%
R217 10k 1%
TP201 VCC_1V8
TP202 PWRON
PWR_FLAG
VCC3V3_SYS
VCC_3V3
VCC3V3_SD

Ingressi (reference AIoT): VCC1 22 uF, VCC2/VCC3/VCC4/VCC9 10 uF, VCC5-VCC8 1 uF, tutti vicino ai pin.

BUCK: induttori 0.47 uH 2520 (DCR < 50 mohm, datasheet: 0.47 uH/4 A BUCK1/2/5, 3 A BUCK3/4). FB1/FB2/FB4 sulle uscite (VDD_LOGIC/GPU/NPU), come il reference.

BUCK3 VCC_DDR = 0.8 V x (1 + 62k/120k) = 1.21 V per DDR4 (tabella del reference: DDR4 1.21 V con 62K 1%).

Gestione: VREF 1 uF, VCC_1P8A/1P8D 1 uF, VCC_RTC 1 uF, quarzo 32.768 kHz Y201 con 2x 22 pF (reference). SDA/SCL 22R serie verso I2C0.

BUCK5 VCC_1V8 (1.8 V, seq 2) libero: la VCCQ dell'eMMC e VCCIO2 restano sull'AP2112K-1.8 come richiesto. Test point TP201.

RESETB -> RESETn (nPOR del SoC) con 10k a VCC3V3_PMU e 100 nF. CLK32K (open drain) -> 0R -> CLK32K: CLK32K_IN del SoC (GPIO0_B0) e LPO dell'AP6256, 10k a VCC3V3_PMU.

PWRON con 100 nF e test point (accensione a mano). VDC: partitore da 5V_SYS, la PMIC parte da sola quando arriva l'alimentazione.

LDO e switch (reference AIoT): 1 uF su ogni uscita, LDO4 4.7 uF, SWOUT1 10 uF. LDO1/LDO4/LDO9 (IMAGE, ACODEC) non usati: spenti di default, solo il condensatore.

LDO2 VDDA_0V9, LDO3 VDDA0V9_PMU, LDO5 VCCIO_SD (VCCIO3), LDO6 VCC3V3_PMU (PMUIO1/2), LDO7 VCCA_1V8, LDO8 VCCA1V8_PMU, SWOUT1 VCC_3V3 (VCCIO4-7 e periferiche), SWOUT2 VCC3V3_SD (microSD).

Codec non usato (nota del reference): pin 14, 15, 16, 17, 19, 40 a VSS; 18, 32, 34-39, 41-43 flottanti. VCC_SPK_HP (33) da VCC3V3_SYS: il datasheet chiede VCC_RTC (45) = tensione piu' alta dell'RK809.

VCC_RTC (45) a VCC3V3_SYS con 1 uF (fig. 1-2 del datasheet, senza batteria). VDC (61) da 5V_SYS con 100k/22k = 0.9 V (>0.55 V: accensione al fronte di salita). BATDIV/SNSP/SNSN a GND. EXT_EN libero.

VDD_CPU: TCS4525 (datasheet TCS Ver.1.0) come reference AIoT pag. 23 e ROCK 3A: VIN da 5V_SYS, I2C0 0x1C, VSEL da PMIC_SLEEP_H, EN da VCC3V3_PMU con 51k (parte dopo LDO6).

FB (A4) = VDD_CPU_COM (sfera M15 del SoC, sense remoto) con 100R verso VDD_CPU. Induttore 0.33 uH (datasheet: 0.33 o 0.47 uH), Isat > 8 A (picco 8 A). PGND/PG (B3) a GND.

VCC3V3_SYS: TPS563201 da 5V_SYS, 3.3 V / 3 A, sempre acceso (EN 100k a 5V_SYS). Alimenta RK809-5 (VCC1-9), FUSB302B e i due LDO. Le periferiche stanno su VCC_3V3 (SWOUT1).

VCC1V8_EMMC: AP2112K-1.8TRG1 (LCSC C176944) da VCC3V3_SYS per VCCQ dell'eMMC e VCCIO2 (FLASH_VOL_SEL alto). EN da VCC3V3_PMU: sale con la sequenza della PMIC.

VCC2V5_DDR (VPP della DDR4): AP2112K-2.5TRG1 da VCC3V3_SYS, EN da VCC3V3_PMU (seq 2, prima di VCC_DDR seq 3: VPP >= VDD come chiede JEDEC). Uscita 2x 4.7 uF + 2x 100 nF (reference).

Vout = 0.768 V x (1 + 33.2k/10k) = 3.32 V. Carico stimato 2.5-3 A di picco (bucks RK809 + SWOUT1/2): al limite dei 3 A, vedi punti aperti.

Sheet: /PMIC/
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MEMORIE: DDR4 x16 (1 chip, point–to–point, niente VTT), eMMC 8 bit su EMMC (VCCIO2 1.8 V), microSD su SDMMC0 (VCCIO3). Serie 4xx.

U401 Samsung K4AAG165WA–BCWE (LCSC C2920140, 16 Gbit = 2 GB, x16, DDR4–3200): VDD/VDDQ = VCC_DDR 1.2 V (BUCK3), VPP = VCC2V5_DDR (AP2112K–2.5). Collegata al canale A del PHY dell'RK3568.

DQ/DM/DQS secondo il pinout Rockchip (DDR_DQx_A –> DDR4_DQL/DQU_A): l'ordine dei bit nel byte e' quello del template Rockchip, i CA non si possono scambiare (Hardware Design Guide 2.1.7).

ZQ 240R 1% a GND, TEN e PAR a GND, ALERT_n libero, T7 (NC/A17) libero (reference Rockchip DDR4 16bit SDP). RESET_n con 1 nF DNP (anti–ESD, Hardware Design Guide).

VREFCA da DDR_VREFOUT del SoC (P8, 0.6 V di default) con 0R; 2x 1 nF vicino alla DDR4 (valore da non cambiare, Hardware Design Guide). Partitore 1k/1k DNP come alternativa.

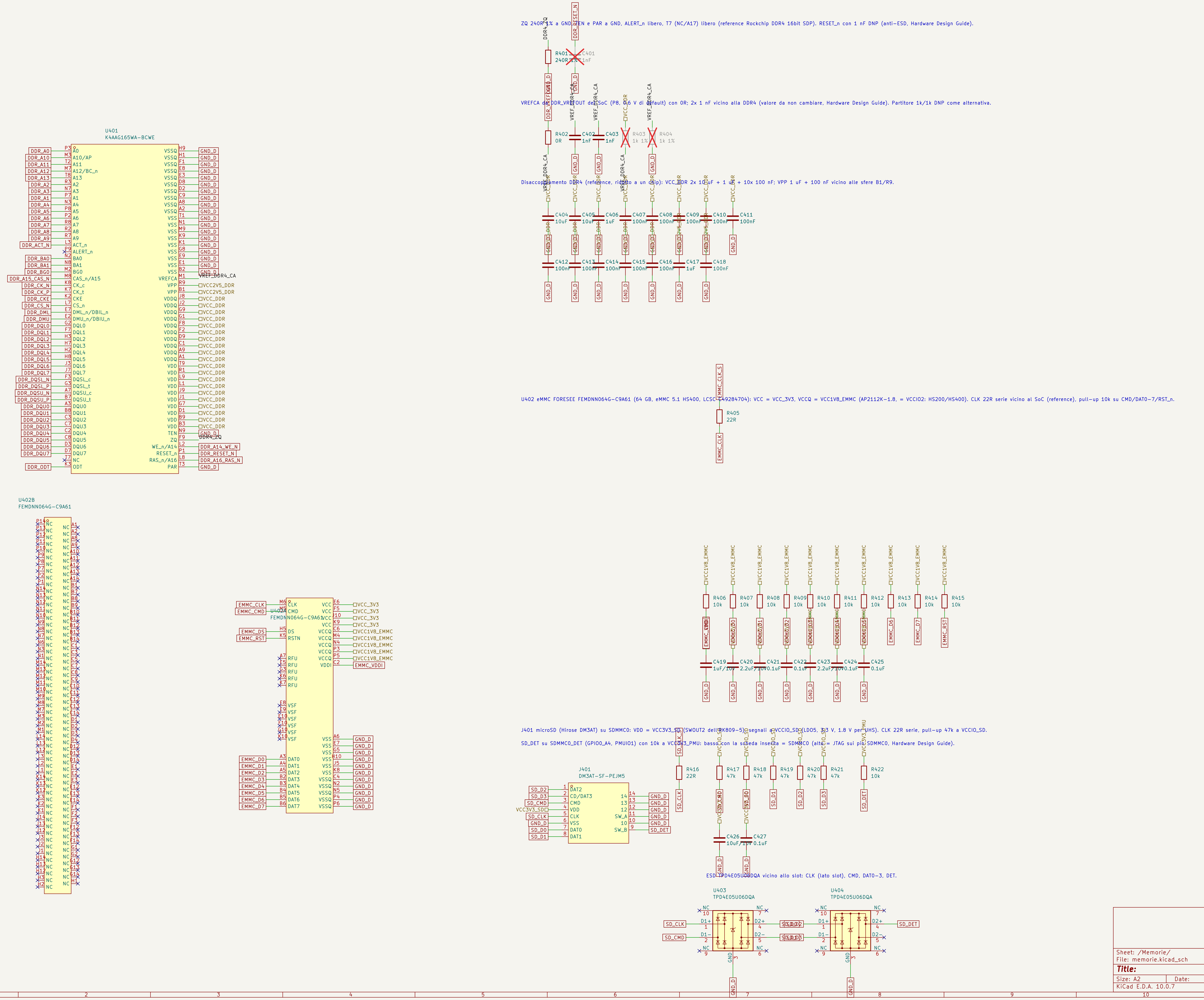
Disaccoppiamento DDR4 (reference, ridotto a un chip): VCC_DDR 2x 10 uF + 1 uF + 10x 100 nF; VPP 1 uF + 100 nF vicino alle sfere B1/R9.

U402 eMMC FORESEE FEMDNN064G–C9A61 (64 GB, eMMC 5.1 HS400, LCSC C49284704): VCC = VCC_3V3, VCCQ = VCC1V8_EMMC (AP2112K–1.8, = VCCIO2: HS200/HS400). CLK 22R serie vicino al SoC (reference), pull–up 10k su CMD/DAT0–7/RST_n.

J401 microSD (Hirose DM3AT) su SDMMC0: VDD = VCC3V3_SD (SWOUT2 dell'RK809–5), segnali a VCCIO_SD (LDO5, 3.3 V, 1.8 V per UHS). CLK 22R serie, pull–up 47k a VCCIO_SD.

SD_DET su SDMMC0_DET (GPIO0_A4, PMUIO1) con 10k a VCC3V3_PMU: basso con la scheda inserita = SDMMC0 (alto = JTAG sui pin SDMMC0, Hardware Design Guide).

ESD TPD4E05U06DQA vicino allo slot: CLK (lato slot), CMD, DAT0–3, DET.


U401
K4AAG165WA–BCWE
VREF_DDR4_CA
DDR4_ZQ
R401 240R 1%
C401 1nF
DDR_RESET_N
R402 0R
C402 1nF
C403 1nF
R403 1k 1%
R404 1k 1%
C404 10uF
C405 10uF
C406 1uF
C407 100nF
C408 100nF
C409 100nF
C410 100nF
C411 100nF
C412 100nF
C413 100nF
C414 100nF
C415 100nF
C416 100nF
C417 1uF
C418 100nF
U402B
FEMDNN064G–C9A61
U402A
FEMDNN064G–C9A61
EMMC_CLK_S
R405 22R
EMMC_CLK
R406 10k
R407 10k
R408 10k
R409 10k
R410 10k
R411 10k
R412 10k
R413 10k
R414 10k
R415 10k
C419 1uF/10V
C420 2.2uF/10V
C421 0.1uF
C422 0.1uF
C423 2.2uF/10V
C424 0.1uF
C425 0.1uF
J401
DM3AT–SF–PEJM5
R416 22R
R417 47k
R418 47k
R419 47k
R420 47k
R421 47k
R422 10k
C426 10uF/10V
C427 0.1uF
U403
TPD4E05U06DQA
U404
TPD4E05U06DQA

Sheet: /Memorie/
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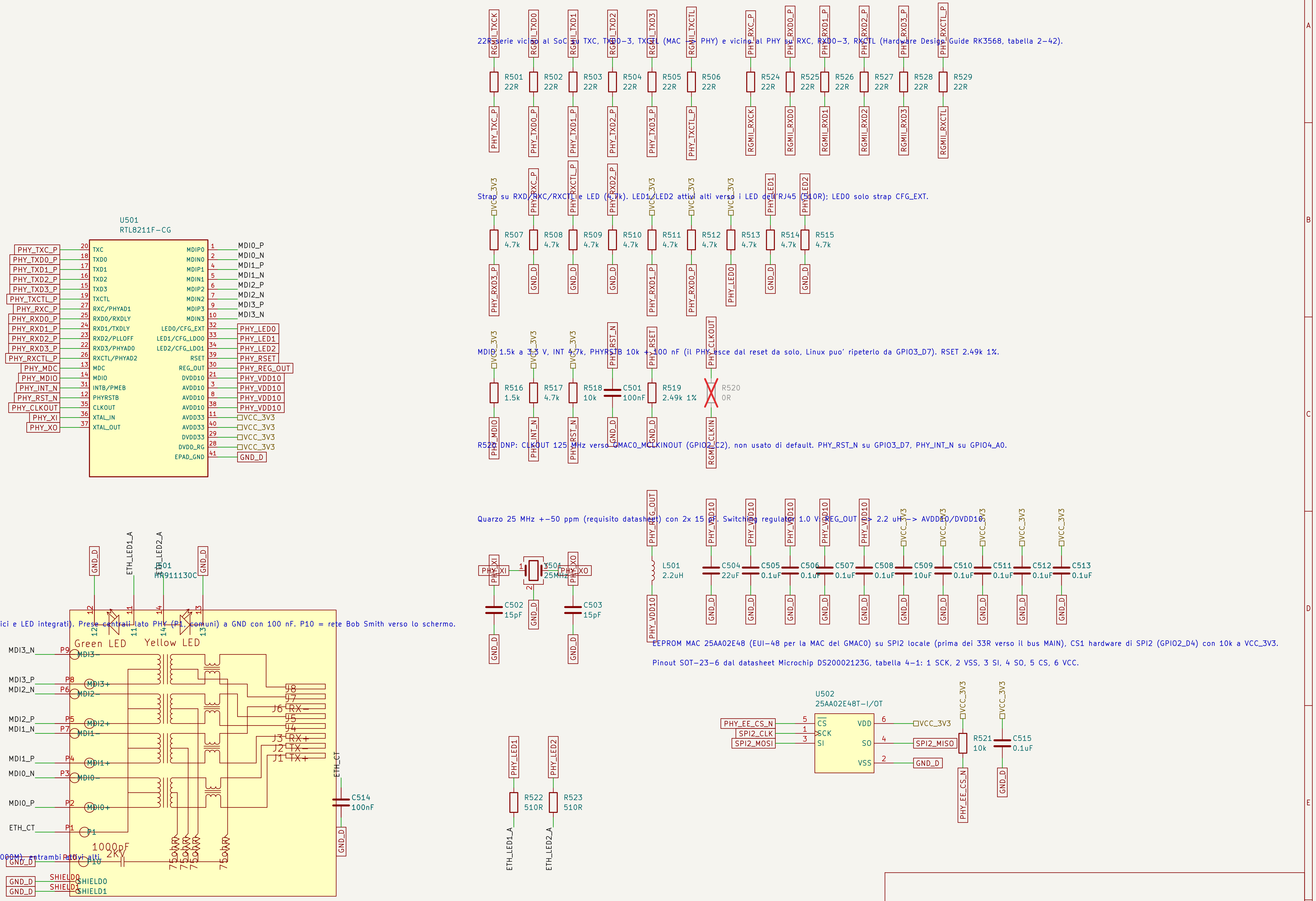
RETE: PHY Realtek RTL8211F-CG (datasheet rev. 1.7) su GMAC0 RGMII dell'RK3568 (VCCIO4 3.3 V), alimentato da VCC_3V3 (SWOUT1). Serie 5xx. Net RGMII_*/PHY_* globali verso il foglio CPU.

Strap (4.7k): PHYAD = 001, PLLOFF = 0, TXDLY = 1 e RXDLY = 1 (rgmii-id, ritardi nel PHY), CFG_EXT = 1 con CFG_LDO = 00: pad RGMII alimentati da fuori a 3.3 V.


U501
RTL8211F-CG
PHY_TXC_P
PHY_TXD0_P
PHY_TXD1_P
PHY_TXD2_P
PHY_TXD3_P
PHY_TXCTL_P
PHY_RXC_P
PHY_RXD0_P
PHY_RXD1_P
PHY_RXD2_P
PHY_RXD3_P
PHY_RXCTL_P
PHY_MDC
PHY_MDIO
PHY_INT_N
PHY_RST_N
PHY_CLKOUT
PHY_XI
PHY_XO
TXC
TXD0
TXD1
TXD2
TXD3
TXCTL
RXC/PHYAD1
RXD0/RXDLY
RXD1/TXDLY
RXD2/PLLOFF
RXD3/PHYAD0
RXCTL/PHYAD2
MDC
MDIO
INTB/PMEB
PHYRSTB
CLKOUT
XTAL_IN
XTAL_OUT
MDIP0
MDIN0
MDIP1
MDIN1
MDIP2
MDIN2
MDIP3
MDIN3
LED0/CFG_EXT
LED1/CFG_LDO0
LED2/CFG_LDO1
RSET
REG_OUT
DVDD10
AVDD10
AVDD33
DVDD33
DVDD_RG
EPAD_GND
MDI0_P
MDI0_N
MDI1_P
MDI1_N
MDI2_P
MDI2_N
MDI3_P
MDI3_N
PHY_LED0
PHY_LED1
PHY_LED2
PHY_RSET
PHY_REG_OUT
PHY_VDD10
VCC_3V3
GND_D
RGMII_TXCK
RGMII_TXD0
RGMII_TXD1
RGMII_TXD2
RGMII_TXD3
RGMII_TXCTL
RGMII_RXCK
RGMII_RXD0
RGMII_RXD1
RGMII_RXD2
RGMII_RXD3
RGMII_RXCTL
R501 22R
R502 22R
R503 22R
R504 22R
R505 22R
R506 22R
R524 22R
R525 22R
R526 22R
R527 22R
R528 22R
R529 22R
R507 4.7k
R508 4.7k
R509 4.7k
R510 4.7k
R511 4.7k
R512 4.7k
R513 4.7k
R514 4.7k
R515 4.7k
R516 1.5k
R517 4.7k
R518 10k
C501 100nF
R519 2.49k 1%
R520 0R
RGMII_CLKIN
Y501 25MHz
C502 15pF
C503 15pF
L501 2.2uH
C504 22uF
C505 0.1uF
C506 0.1uF
C507 0.1uF
C508 0.1uF
C509 10uF
C510 0.1uF
C511 0.1uF
C512 0.1uF
C513 0.1uF
J501
HR911130C
Green LED
Yellow LED
ETH_LED1_A
ETH_LED2_A
MDI3-
MDI3+
MDI2-
MDI2+
MDI1-
MDI1+
MDI0-
MDI0+
P1
P10
SHIELD0
SHIELD1
1000pF 2KV
75ohm
J8
J7
J6 RX-
J5
J4
J3 RX+
J2 TX-
J1 TX+
ETH_CT
C514 100nF
R522 510R
R523 510R
U502
25AA02E48T-I/OT
PHY_EE_CS_N
SPI2_CLK
SPI2_MOSI
SPI2_MISO
CS
SCK
SI
VDD
SO
VSS
R521 10k
C515 0.1uF

22R serie vicino al SoC su TXC, TXD0–3, TXCTL (MAC -> PHY) e vicino al PHY su RXC, RXD0–3, RXCTL (Hardware Design Guide RK3568, tabella 2–42).

Strap su RXD/RXC/RXCTL e LED (4.7k). LED1/LED2 attivi alti verso i LED dell'RJ45 (510R); LED0 solo strap CFG_EXT.

MDIO 1.5k a 3.3 V, INT 4.7k, PHYRSTB 10k + 100 nF (il PHY esce dal reset da solo, Linux puo' ripeterlo da GPIO3_D7). RSET 2.49k 1%.

R520 DNP: CLKOUT 125 MHz verso GMAC0_MCLKINOUT (GPIO2_C2), non usato di default. PHY_RST_N su GPIO3_D7, PHY_INT_N su GPIO4_A0.

Quarzo 25 MHz +–50 ppm (requisito datasheet) con 2x 15 pF. Switching regulator 1.0 V: REG_OUT –> 2.2 uH –> AVDD10/DVDD10.

RJ45 HanRun HR911130C (100/1000Base–T, magnetici e LED integrati). Prese centrali lato PHY (P1, comuni) a GND con 100 nF. P10 = rete Bob Smith verso lo schermo.

EEPROM MAC 25AA02E48 (EUI–48 per la MAC del GMAC0) su SPI2 locale (prima dei 33R verso il bus MAIN), CS1 hardware di SPI2 (GPIO2_D4) con 10k a VCC_3V3.

Pinout SOT–23–6 dal datasheet Microchip DS20002123G, tabella 4–1: 1 SCK, 2 VSS, 3 SI, 4 SO, 5 CS, 6 VCC.

LED verde = LED1 del PHY (100M), giallo = LED2 (1000M), entrambi attivi alti.

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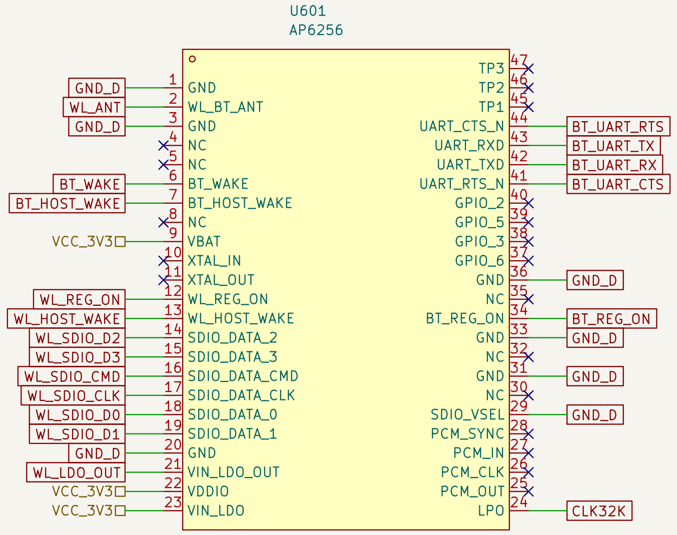
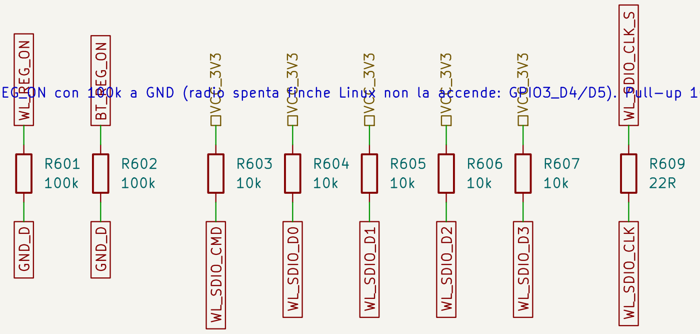
RADIO: AMPAK AP6256 (Wi-Fi 5 + BT 5.0) su SDMMC2_M0 (SDIO, VCCIO6), HCI su UART8_M0 con RTS/CTS (VCCIO4), REG_ON e wake su GPIO. Serie 6xx. Net WL_*/BT_*/CLK32K globali.

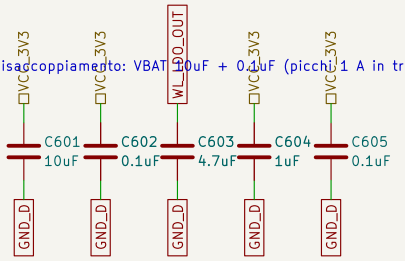
VDDIO = VCC_3V3 3.3 V (= VCCIO6/VCCIO4), SDIO 2.0 fino a 50 MHz. VBAT e VIN_LDO da VCC_3V3. LPO 32.768 kHz da CLK32K dell'RK809-5 (quarzo Y201).


U601
AP6256
GND_D 1 GND
WL_ANT 2 WL_BT_ANT
GND_D 3 GND
4 NC
5 NC
BT_WAKE 6 BT_WAKE
BT_HOST_WAKE 7 BT_HOST_WAKE
8 NC
VCC_3V3 9 VBAT
10 XTAL_IN
11 XTAL_OUT
WL_REG_ON 12 WL_REG_ON
WL_HOST_WAKE 13 WL_HOST_WAKE
WL_SDIO_D2 14 SDIO_DATA_2
WL_SDIO_D3 15 SDIO_DATA_3
WL_SDIO_CMD 16 SDIO_DATA_CMD
WL_SDIO_CLK 17 SDIO_DATA_CLK
WL_SDIO_D0 18 SDIO_DATA_0
WL_SDIO_D1 19 SDIO_DATA_1
GND_D 20 GND
WL_LDO_OUT 21 VIN_LDO_OUT
VCC_3V3 22 VDDIO
VCC_3V3 23 VIN_LDO
TP3 47
TP2 46
TP1 45
UART_CTS_N 44 BT_UART_RTS
UART_RXD 43 BT_UART_TX
UART_TXD 42 BT_UART_RX
UART_RTS_N 41 BT_UART_CTS
GPIO_2 40
GPIO_5 39
GPIO_3 38
GPIO_6 37
GND 36 GND_D
NC 35
BT_REG_ON 34 BT_REG_ON
GND 33 GND_D
NC 32
GND 31 GND_D
NC 30
SDIO_VSEL 29 GND_D
PCM_SYNC 28
PCM_IN 27
PCM_CLK 26
PCM_OUT 25
LPO 24 CLK32K

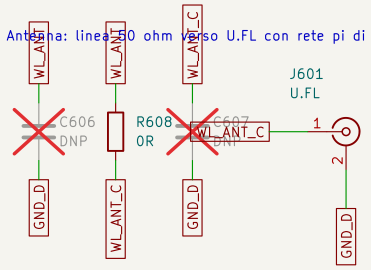
REG_ON con 100k a GND (radio spenta finche Linux non la accende: GPIO3_D4/D5). Pull-up 10k su CMD e DAT0-3 a VCC_3V3. CLK 22R serie vicino al SoC.


WL_REG_ON
BT_REG_ON
VCC_3V3
VCC_3V3
VCC_3V3
VCC_3V3
VCC_3V3
WL_SDIO_CLK_S
R601
100k
R602
100k
R603
10k
R604
10k
R605
10k
R606
10k
R607
10k
R609
22R
GND_D
GND_D
WL_SDIO_CMD
WL_SDIO_D0
WL_SDIO_D1
WL_SDIO_D2
WL_SDIO_D3
WL_SDIO_CLK

Disaccoppiamento: VBAT 10uF + 0.1uF (picchi 1 A in trasmissione), VIN_LDO_OUT 4.7uF, VDDIO 1uF + 0.1uF.


VCC_3V3
VCC_3V3
WL_LDO_OUT
VCC_3V3
VCC_3V3
C601
10uF
C602
0.1uF
C603
4.7uF
C604
1uF
C605
0.1uF
GND_D
GND_D
GND_D
GND_D
GND_D

Antenna: linea 50 ohm verso U.FL con rete pi di adattamento (0R + 2x DNP) da tarare.


WL_ANT
WL_ANT
WL_ANT_C
J601
U.FL
C606
DNP
R608
0R
C607
DNP
WL_ANT_C
1
2
GND_D
WL_ANT_C
GND_D
GND_D

UART incrociata: TXD del modulo -> BT_UART_RX (UART8_RX, GPIO2_C6), RXD <- BT_UART_TX (GPIO2_C5), RTS_N -> BT_UART_CTS (GPIO2_B2), CTS_N <- BT_UART_RTS (GPIO2_B1).

PCM (25-28) non collegati: niente audio Bluetooth. XTAL_IN/OUT liberi (quarzo interno). GPIO e TP liberi.

Sheet: /Radio/
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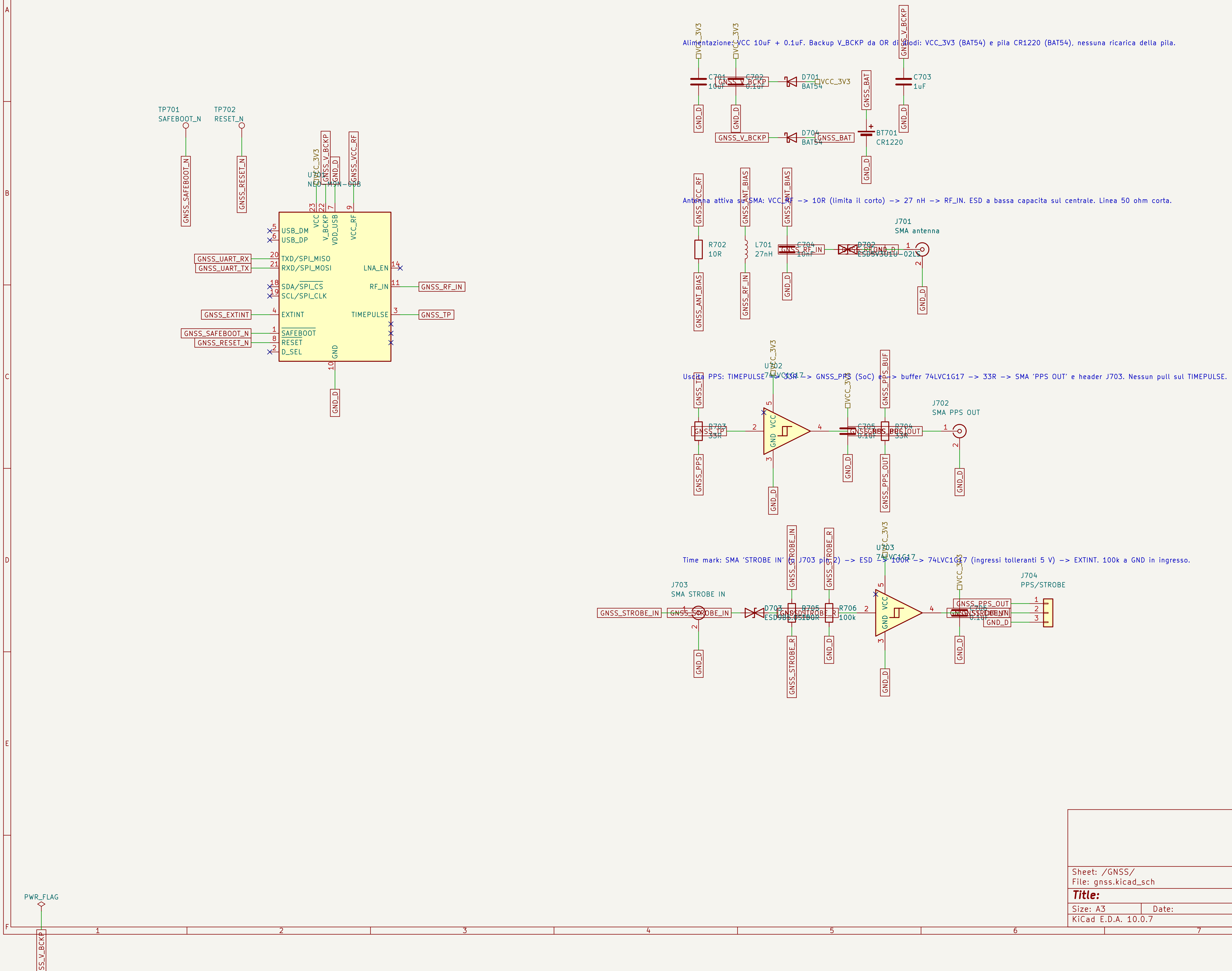
GNSS: u-blox NEO-M9N-00B (formato NEO, stesso footprint del NEO-F10T). Pin dal datasheet UBX-19014285 R08, tabella 9 (= simbolo NEO-M9N di KiCad). Serie 7xx.

UART3_M1: TXD del modulo -> GNSS_UART_RX (GPIO3_C0), RXD <- GNSS_UART_TX (GPIO3_B7). TIMEPULSE -> 33R -> GNSS_PPS (GPIO3_A5). EXTINT (pin 4, PIO7, time mark TIM-TM2) letto anche da GPIO3_A6 (solo ingresso).

D_SEL (pin 2) aperto = UART + I2C. V_USB (pin 7) a GND: USB non usata (da confermare sull'integration manual). LNA_EN e riservati liberi.



Alimentazione: VCC 10uF + 0.1uF. Backup V_BCKP da OR di diodi: VCC_3V3 (BAT54) e pila CR1220 (BAT54), nessuna ricarica della pila.

Antenna attiva su SMA: VCC_RF -> 10R (limita il corto) -> 27 nH -> RF_IN. ESD a bassa capacita sul centrale. Linea 50 ohm corta.

Uscita PPS: TIMEPULSE -> 33R -> GNSS_PPS (SoC) e -> buffer 74LVC1G17 -> 33R -> SMA 'PPS OUT' e header J703. Nessun pull sul TIMEPULSE.

Time mark: SMA 'STROBE IN' (o J703 pin 2) -> ESD -> 100R -> 74LVC1G17 (ingressi tolleranti 5 V) -> EXTINT. 100k a GND in ingresso.

TP701
SAFEBOOT_N
TP702
RESET_N
GNSS_SAFEBOOT_N
GNSS_RESET_N
U701
NEO-M9N-00B
VCC_3V3
GNSS_V_BCKP
GND_D
GNSS_VCC_RF
VCC
V_BCKP
VDD_USB
VCC_RF
USB_DM
USB_DP
TXD/SPI_MISO
RXD/SPI_MOSI
SDA/SPI_CS
SCL/SPI_CLK
EXTINT
SAFEBOOT
RESET
D_SEL
GND
LNA_EN
RF_IN
TIMEPULSE
GNSS_UART_RX
GNSS_UART_TX
GNSS_EXTINT
GNSS_RF_IN
GNSS_TP
C701
10uF
C702
0.1uF
D701
BAT54
VCC_3V3
C703
1uF
D704
BAT54
GNSS_BAT
BT701
CR1220
GNSS_ANT_BIAS
R702
10R
L701
27nH
C704
10nF
D702
ESD5V3U1U-02LS
J701
SMA antenna
U702
74LVC1G17
R703
33R
C705
0.1uF
R704
33R
GNSS_PPS
GNSS_PPS_BUF
GNSS_PPS_OUT
J702
SMA PPS OUT
J703
SMA STROBE IN
GNSS_STROBE_IN
D703
ESD9B5.0ST5G
R705
100R
GNSS_STROBE_R
R706
100k
U703
74LVC1G17
C706
0.1uF
J704
PPS/STROBE
PWR_FLAG

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Rev:

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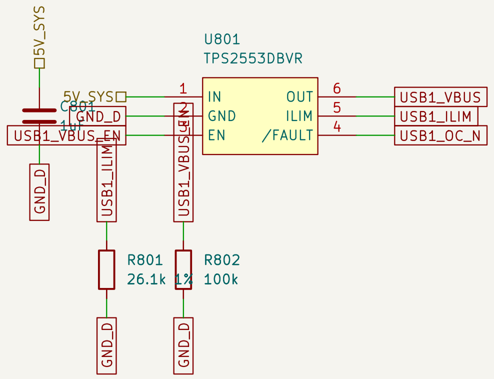
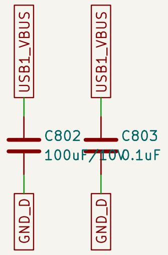
USB HOST: USB3_HOST1_DP/DM dell'RK3568 (parte USB 2.0, alimentata da USB3_AVDD) su USB−C (16 pin, HRO TYPE−C−31−M−12), solo host. Serie 8xx. Net USB1_* globali.

La porta non alimenta mai la scheda: VBUS esce solo dal load switch. Rp fissi da 22k a 5V_SYS su CC1/CC2 = sorgente 1.5 A (Type−C), niente controller PD.

TPS2553: EN attivo alto da GPIO0_A6 con 100k a GND, FAULT open−drain su GPIO0_B6 (pull−up 10k a VCC3V3_PMU sul foglio CPU).

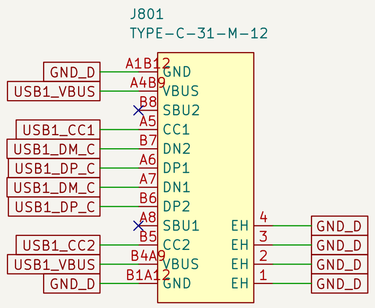
RILIM 26.1k 1% = 0.91 / 0.99 / 1.08 A (min/nom/max, tabella 2 del datasheet).

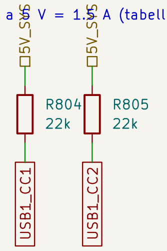

5V_SYS
U801
TPS2553DBVR
5V_SYS
1
IN
OUT
6
USB1_VBUS
C801
1uF
GND_D
2
GND
ILIM
5
USB1_ILIM
USB1_VBUS_EN
3
EN
/FAULT
4
USB1_OC_N
GND_D
USB1_ILIM
USB1_VBUS_EN
R801
26.1k 1%
R802
100k
GND_D
GND_D


USB1_VBUS
USB1_VBUS
C802
100uF/10V
C803
0.1uF
GND_D
GND_D

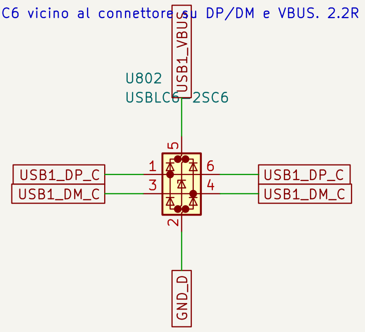
J801 USB−C: DP/DM collegati su entrambi i lati (A6/B6, A7/B7), SBU non connessi, schermo a GND.

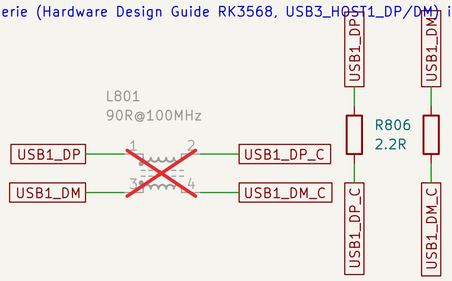
Rp 22k a 5 V = 1.5 A (tabella Type−C). Il TPS2553 limita comunque a circa 1 A.


J801
TYPE−C−31−M−12
GND_D
A1B12
GND
USB1_VBUS
A4B9
VBUS
B8
SBU2
USB1_CC1
A5
CC1
USB1_DM_C
B7
DN2
USB1_DP_C
A6
DP1
USB1_DM_C
A7
DN1
USB1_DP_C
B6
DP2
A8
SBU1
EH
4
GND_D
USB1_CC2
B5
CC2
EH
3
GND_D
USB1_VBUS
B4A9
VBUS
EH
2
GND_D
GND_D
B1A12
GND
EH
1
GND_D


5V_SYS
5V_SYS
R804
22k
R805
22k
USB1_CC1
USB1_CC2

ESD USBLC6−2SC6 vicino al connettore su DP/DM e VBUS. 2.2R in serie (Hardware Design Guide RK3568, USB3_HOST1_DP/DM) in parallelo al choke di modo comune DNP.


USB1_VBUS
U802
USBLC6−2SC6
5
USB1_DP_C
1
6
USB1_DP_C
USB1_DM_C
3
4
USB1_DM_C
2
GND_D


L801
90R@100MHz
USB1_DP
1
2
USB1_DP_C
USB1_DM
3
4
USB1_DM_C
USB1_DP
USB1_DM
R806
2.2R
R807
2.2R
USB1_DP_C
USB1_DM_C

Sheet: /USB host/
File: usb.kicad_sch

Title:

Size: A3 | Date: | **Rev:**
KiCad E.D.A. 10.0.7 | Id: 10/10